F78PA Probability & Statistics A

Tutorial 6

1. There are two types, A and B, of rechargeable batteries in a bin. When in use, a type A battery lasts an exponentially distributed time (in hours) with rate $\lambda_A = \frac{1}{20}$, and a type B battery lasts an exponentially distributed time (in hours) with rate $\lambda_B = \frac{1}{30}$. A battery that is random chosen from the bin will be a type A battery with probability 2/3 and a type B battery with probability 1/3.

   (a) Given that a randomly chosen battery is still working after 20 hours, what is the probability that it is a type A battery?

   (b) Given that a random chosen battery is still working after 20 hours, what is the probability that it will still be operating after another 10 hours?

2. Suppose that $X_1, X_2, ..., X_n$ are *independent* random variables with distribution functions $F_1(x), F_2(x), ..., F_n(x)$ respectively. Let $M = \min\{X_1, X_2, ..., X_n\}$.

   (a) Show that for any $x \in \mathbb{R}$,

   $$\mathbb{P}\{M > x\} = \mathbb{P}\{X_1 > x\}\mathbb{P}\{X_2 > x\} \cdots \mathbb{P}\{X_n > x\}.$$

   (b) Let $F_M(x)$ denote the distribution function for M. Use part (a) above to show that for any $x \in \mathbb{R}$,

   $$F_M(x) = 1 - (1 - F_1(x))(1 - F_2(x)) \cdots (1 - F_n(x)).$$

   (c) Sam wants to take a nap for 1 hour and then get up and go to his tutorial class. He is a heavy sleeper, so he decides to set 5 alarm clocks to wake him up. Let T_i denote the time that the ith clock rings ($i = 1, 2, ..., 5$), and suppose that for each $i = 1, 2, ..., 5$, the time T_i is exponentially distributed with mean 1 hour and the times $T_1, T_2, ..., T_5$ are independent random variables
   Determine the distribution of the time $M = \min\{T_1, T_2, ..., T_n\}$ that the first clock goes off and determine $\mathbb{E}(M)$.

3. Suppose that the random variables X and Y have joint distribution function $F_{X,Y}(x, y)$. Show that for any real numbers $a < b$ and $c < d$,

   $$\mathbb{P}\{a < X \leq b, c < Y \leq d\} = F_{X,Y}(b, d) - F_{X,Y}(b, c) - F_{X,Y}(a, d) + F_{X,Y}(a, c).$$

4. Two random variables X and Y have joint probability mass function given by the following:

	$y = 0$	1	2	3
$x = 0$	0	3/56	6/56	1/56
1	3/56	18/56	9/56	0
2	6/56	9/56	0	0
3	1/56	0	0	0

   Find:

   (a) the probability $\mathbb{P}(X < Y)$.

   (b) the marginal probability mass function for X and determine:
      i. $\mathbb{P}(X \geq 2)$, and
      ii. $E(X)$, the expected value of X;

   (c) the marginal probability mass function for Y and determine:
      i. $\mathbb{P}(Y \leq 1)$, and

ii. $E(Y)$, the expected value of Y;

(d) the probability mass function for $Z = X - Y$ and determine:
   i. $\mathbb{P}(-1 \leq Z \leq 1)$, and
   ii. $E(Z)$, the expected value of Z;
   iii. σ_Z, the standard deviation of Z.

(e) the conditional probability mass function of X *given* $Y = 0$ and determine:
   i. $\mathbb{P}(X \geq 2 \mid Y = 0)$, and
   ii. $E(X \mid Y = 0)$, the conditional expectation of X *given* $Y = 0$;

(f) the conditional probability mass function of Y *given* $X = 1$ and determine:
   i. $\mathbb{P}(Y \leq 1 \mid X = 1)$, and
   ii. $E(Y \mid X = 1)$, the conditional expectation of Y *given* $X = 1$.

(g) Are the variables X and Y independent? Give reasons for your answer.

5. Suppose that X is an exponential random variable with parameter $\lambda > 0$.

(a) **Memoryless property** Show that for any $t, s > 0$,

$$\mathbb{P}\{X > s + t \mid X > t\} = \mathbb{P}\{X > s\}.$$

Remark: This shows that exponential random variables have the *memoryless property*. When X is interpreted as the waiting time for an event to occur, this relation implies that, if X is conditioned on a failure to observe the event over some initial period of time t, the distribution of the remaining waiting time is the same as the original unconditional distribution.
Exponential random variables are the *only* continuous random variables with the memoryless property.

(b) Suppose that the time, T, that it takes an engineer to repair a machine is exponentially distributed with mean 2 hours. Determine
   i. the probability it will take more than 2 hours to repair the machine,
   ii. the probability that it will take more than 6 hours to repair the machine *given* that the engineer has already spent 4 hours trying to repair the machine.

Exam Style Question

(a) Suppose that X_1 and X_2 are both independent and exponentially distributed with mean λ. Let $T = \max(X_1, X_2)$.

i) Find the cumulative distribution function, F_T, of T, and the probability density function, f_T, of T. [6 marks]

ii) Find the expected value, $\mathbb{E}[T]$, of T. [5 marks]

(b) A lamp has two LED light bulbs. It will operate if at least one of these bulbs are functioning. The lifetime of each bulb is exponentially distributed with mean of 1 year and the lifetime of the two bulbs are independent. Find the probability that the lamp is functioning after 2 years. [2 marks]

[Total 13 marks]